

MATTHEW BERGER

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Founding engineer with 11 years Rust experience (pre-1.0 early adopter), including 5+ years in production systems. Built food assembly robotics control suite from zero to production in 22 months, securing \$10M+ investment from Chipotle and Cava.

WORK EXPERIENCE

Hyphen (Culinary Robotics)

Remote

Senior Staff Software Engineer (Founding Engineer)

January 2026 - May 2026

- Architected and delivered Hyphen Studio, a Dioxus desktop application for operator makeline configuration
- Replaced TypeScript/GraphQL tablet portal with a pure Rust Leptos implementation connecting directly to the IPC broker
- Built Rust printer driver and label visualization system for Zebra printers enabling professional branded labels

Hyphen (Culinary Robotics)

Remote

Staff Software Engineer (Founding Engineer)

September 2022 - January 2026

- Founded food assembly robotics controls suite as first engineer, driving technical pivot from PLC to Rust architecture
- Merged **1,000+ pull requests** over 4 years as sole architect across firmware, controls, IPC, and cloud infrastructure
- Achieved complete product delivery in **22 months** securing **\$10M+ investment** from Chipotle and Cava
- Transformed team of TypeScript/React developers into proficient Rust systems programmers through hands-on mentorship, establishing architectural patterns, and pair programming
- Engineered full technology stack from bare-metal firmware (RP2040/Embassy) to distributed robotics control system and cloud infrastructure, including custom Yocto Linux distribution
- Built core async message broker coordinating distributed processes on industrial PC with direct networking to embedded boards, handling **1M+ msgs/sec** with **<1ms latency**, zero message loss over **3 years** of constant use
- Designed deterministic layered configuration system enabling unlimited fleet scaling
- Created Hyphen Explorer tool achieving **100% adoption** across all software engineers, ops technicians, and culinary operators for real-time IPC visualization, debugging, and system control

Hyphen (Culinary Robotics)

Remote

Senior Software Engineer

July 2021 - September 2022

- Diagnosed critical scaling limitations in legacy PLC system preventing growth beyond prototype
- Led technical pivot from \$500K PLC system to embedded Rust architecture
- Pioneered RP2040 firmware with Embassy-rs, creating first Rust TMC5160 motor and AS5048A encoder drivers proving Rust could meet real-time control loop requirements
- Built early AWS infrastructure with Pulumi and greengrass IoT

Sierra Nevada Corporation

Englewood, CO

Software Engineer III

May 2020 - July 2021

- Developed C++/Rust aerospace imaging system (5GB/sec) and simulator saving 3 months on delivery timeline

Scientific Games

Reno, NV

Software Engineer

July 2019 - May 2020

- Resolved critical performance and behavioral defects in a Unity-based casino-gaming platform

Hamilton Company

Reno, NV

Software Engineer

January 2018 - July 2019

Software Engineering Intern

October 2014 - December 2017

- Built safety-critical software for liquid-handling medical robots
- Automated gravimetric analysis QA process, saving **1+ week per robot** across production fleet
- Reduced calibration development from **2 months to 2 weeks** through reusable plugin framework
- Architected diagnostic application for largest OEM customer Illumina (**\$2M contract**)

OPEN SOURCE

- **Nightshade**: cross-platform data-oriented game engine in Rust with custom ECS and render graph; runs on Windows, Linux, macOS, and web via WebAssembly
- **freecs**: high-performance ECS library in 1,700 lines of Rust; multi-threaded with Rayon, zero unsafe code, foundation of Nightshade
- **wgpu-example (98 stars, 13 forks)**: minimal Rust + wgpu + egui template without eframe; runs on Windows, Linux, macOS, web, Android, and OpenXR (Quest 3)
- **bamboo**: fast Rust static site generator powering matthewberger.dev
- **enum2 macro family** (enum2str, enum2contract, enum2egui, enum2pos, enum2repr): derive macros for enum-driven design patterns with **230k+ total downloads** on crates.io
- **frost**: dual-backend programming language in Rust with bytecode VM and Cranelift JIT
- **dragonglass**: physically-based 3D rendering engine in Rust and Vulkan (34 stars)

EDUCATION

University of Nevada, Reno BS Computer Science & Engineering, Minor in Mathematics, 2017